

Kassidy Barram, Software Engineer

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LINKS

[LinkedIn](#), [Personal GitHub](#), [Personal Website](#)

PROFILE

Software engineer with 4.5+ years architecting distributed systems and decentralized compute infrastructure, seeking a backend or systems engineering role where I can design and create scalable, resilient, and high-throughput systems and applications.

EMPLOYMENT HISTORY

Sept 2025 — Present

Software Engineer, qBitTensor Labs

Boulder, CO

- Architected a fault-tolerant, distributed validation and scoring platform in Python across 17 independent validators and 250+ miners, enforcing incentive mechanisms across a \$500K prize pool in a permissionless, trust-less compute economy.
- Led development of public and private miner codebases, providing the foundation for participants to submit quantum-inspired solutions to NP-hard optimization problems. The system leverages a modular architecture and inheritance to reduce duplication and ensure maintainability.
- Engineered a security-in-depth Docker-based sandboxed execution environment for untrusted third-party code.
- Built and managed data pipelines and persistence layers using MySQL, SQLite, and SQLAlchemy to support solution tracking, scoring, and network state management.

Oct 2021 — Mar 2025

Software Engineer, Project Sustain

- Designed and built 4 spatio-temporal data visualization web applications in React and deck.gl backed by PostGIS and MongoDB, facilitating high-throughput, low-latency geospatial data rendering across distributed environments.
- Architected a resilient replacement for brittle, failure-prone infrastructure as Kubernetes administrator, enabling continuous deployment, scaling, and scheduling of 24+ distributed services at 99.9% uptime.
- Administered 200 node private cloud.
- Built 2 distributed systems for ML model validation, supporting linear regression and classification across a 75-machine compute network.

May 2019 — Aug 2019

Systems Engineer Intern , CPP Wind Engineering Consultants

- Conducted quantitative testing to evaluate accuracy and frequency response across multiple sensor data collection methods, informing instrument selection and measurement reliability.
- Designed a control station that mixed and regulated the output of tracer gas used in the wind tunnel testing of the air quality from various exhausts produced by buildings.
- Built and assembled city-scape proximity models for data collection in wind tunnels

EDUCATION

Aug 2022 — May 2025

Master of Science, Colorado State University

Fort Collins

- Graduated *Summa cum Laude* (4.0 GPA)
- Awarded Best Paper at 2023 International ACM/IEEE Big Data Conference.
- Published Research in 4 competitive (IEEE/ACM) international computer science conferences.
- Won award at CURC (Celebrating Undergraduate Research and Creativity).

Aug 2017 — May 2022

Bachelor of Science, Colorado State University

Fort Collins

- Graduated *Cum Laude* (3.95GPA)
- Awarded National Science Foundation Research Grant to research distributed systems and big data technologies.
- Participated in research to facilitate communication between Pixhawk 4 Flight Controller and the Inertial Measurement Unit attached to drones using C++.

SKILLS

Java

GitHub

Python

Big Data

C++ / C

Distributed Systems

JavaScript / TypeScript

Docker

Kubernetes

MongoDB

PyTorch / PySpark

gRPC

Hadoop MapReduce

Apache Spark

Tensor Flow

TCP Networking

Test Driven Development

Machine Learning

Artificial Intelligence

RESTful API

Agile

CI/CD Pipelines

Prometheus

Grafana

PUBLICATIONS

- Archimedes: A Framework to Support Distributional Similarity Analysis over Arbitrary Spatiotemporal Scopes at Scale, *IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*, 2025. Paige Hansen, Nathan Orwick, **Kassidy Barram**, Pierce Smith, Jay Breidt, Sangmi Lee Pallickara, Shrideep Pallickara
- Scribe: Enabling Programmatic Interfaces for Explorations Over Voluminous Spatiotemporal Data Collections, *IEEE/ACM International Conference on Big Data Computing, Applications and Technologies (BDCAT)*, 2024. **Kassidy Barram**, Sangmi Lee Pallickara, Shrideep Pallickara
- A Framework for Profiling Spatial Variability in the Performance of Classification Models, *IEEE/ACM International Conference on Big Data Computing, Applications and Technologies (BDCAT)*, 2023. **Best Paper Award**. Menuka Warushavithana, **Kassidy Barram**, Saptashwa Mitra, Caleb Carlson, Sudipto Ghosh, Jay Breidt, Sangmi Lee Pallickara, Shrideep Pallickara
- Resource Efficient Profiling of Spatial Variability in Performance of Regression Models, *IEEE International Conference on Big Data (IEEE BigData)*, 2022. Caleb Carlson, Menuka Warushavithana, Saptashwa Mitra, **Kassidy Barram**, Sudipto Ghosh, Jay Breidt, Sangmi Lee Pallickara, Shrideep Pallickara